

VUTUHARI EVBAIL

DATA SCIENCE

SQL AGGREGATE

FUNCTIONS & OPERATORS

Q1) SELECT DISTINCT * Department
FROM Students;

Department
IT
HR
IT
Finance
HR

Q2) SELECT *
department,
AVG (age) AS average-age
FROM Students;
GROUP BY
department, ~~avg-age~~;

department	avg-age
IT	20
IT	21
HR	22
HR	22
Finance	23

Q3) SELECT department, COUNT(*) AS student count
FROM student
GROUP BY department
HAVING COUNT(*) > 1;

department	Student count
IT	2
HR	2

Q4 SELECT * student id, name, age, department;
FROM Students
WHERE age BETWEEN 21 AND 23;

Student id	name	age	department
3	Charlie	21	IT
2	Bob	22	HR
5	Eve	22	HR
4	Diana	23	Finance

Q5 SELECT * student id, name, age, department;
FROM Students
WHERE Age > 21
AND (Department = 'IT' OR Department = 'HR');

Student id	name	age	department
2	Bob	22	HR
5	Eve	22	HR

Q6 SELECT * department, total credits;
SUM(credits) AS total credits
FROM Courses
GROUP BY department

HAVING

SUM(credits) > 5;

department	total credits
IT	11

67) SELECT * course id, course name, department credits.
FROM courses
WHERE credits != 4;

course id	course name	department	credits
101	SQL Basic	IT	3
104	Excel	Finance	2
105	Statistics	HR	3.

68) SELECT course id, course name credits;
FROM courses
ORDER BY credits DESC
LIMIT 3;

Course id	Course name	Credits
103	Data Science	4
102	python	4
101	SQL Basics	3.

69) SELECT
MAX(grade) AS max-grade,
MIN(grade) AS min grade,
AVG(grade) AS avg grade,
FROM enrollments;

max grade	min agrade	average
90	78	82

10 SELECT

course id,
(COUNT(*) AS enrollment count

FROM enrollment

GROUP BY course id

ORDER BY enrollment count DESC

Course id	enrollment count.
101	1
102	2
103	3
104	4
105	5

11 SELECT * department,

Sum (Salary) AS total salary.

Sum (bonus AS total bonus

FROM employees

GROUP BY department

ORDER BY total salary, total bonus DESC;

department total salary total bonus.

~~IT 12200~~

~~HR 10900~~

Finance

department total salary bonus

Finance 10000 6000

IT 12200 10500

HR 10900 7500

12) SELECT department,
 ROUND(AVG(salary), 2) AS avg_salary
 FROM employees
 GROUP BY department
 HAVING AVG(salary) > 55000
 ORDER BY avg_salary DESC;

department	avg_salary
Finance	70000
IT	62000
IT	60000

13) SELECT employee_id,
 name,
 salary,
 bonus,
 salary + bonus AS total_comp
 FROM employees
 WHERE salary + bonus > 60000
 ORDER BY total_comp DESC.

employee_id	name	salary	bonus	total_compensation
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500
1	Tom	60000	5000	65000
2	Jerry	55000	4000	59000
5	Butch	54000	3500	57500

(14) SELECT

department,
Sum(budget) AS total budget,
ROUND(AVG(budget), 2) AS avg budget
FROM departments
GROUP BY department
HAVING AVG(budget) > 10 000
ORDER BY avg-budget DESC;

department	total budget	avg budget.
IT	270 000	120 000
Finance	80 000	80 000

(15) SELECT project, department, budget

FROM projects

WHERE budget BETWEEN 50000 AND 120 000

AND department != 'Marketing';

project id	project name	department	budget
2	payroll System	Finance	80 000